

Critical Empirical Evaluation of DBN-based NIDS

Final Project Report

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Selected Paper:	Belarbi et al. (2022), Science of Cyber Security
Original Repository:	github.com/othmbela/dbn-based-nids
Dataset Source:	CICIDS2017 Dataset (Canadian Institute for Cybersecurity)

This report documents the rigorous empirical reproduction, pipeline auditing, and performance verification of the Deep Belief Network-based Intrusion Detection System.

1. Executive Summary

This report presents a rigorous empirical reproduction and critical evaluation of the Deep Belief Network (DBN) based Network Intrusion Detection System (NIDS) proposed by Belarbi et al. (2022). Through an exhaustive 16-step methodology pipeline, we audited the original dataset, mitigated class imbalances, eliminated cross-validation leakage, and implemented mathematically robust Explainable AI (XAI).

Our findings fundamentally challenge the authors claims regarding the superiority of deep learning architectures on structured tabular flow data. When evaluated under strict zero-leakage constraints, traditional ensemble methods like Random Forest and XGBoost achieved equal or superior performance metrics (F2, MCC, PR-AUC) while executing orders of magnitude faster than the custom DBN implementation.

2. Source Summary

The source paper "An Intrusion Detection System based on Deep Belief Networks" (Science of Cyber Security, 2022) utilizes the CICIDS2017 dataset. The authors designed a 3-layer DBN trained via Contrastive Divergence and fine-tuned with backpropagation. They claimed state-of-the-art multi-class detection capabilities.

3. Reproduction Methodology

The primary challenge in reproducing the authors findings was identifying and eliminating structural data leakage. Because CICIDS2017 contains massive volumes of exact duplicate flows, random splitting (as used in the original codebase) causes identical network patterns to exist in both the train and test sets, artificially inflating metrics. We implemented a strict Stratified Cross-Validation framework with bootstrapped confidence intervals. Furthermore, data scaling and SMOTE augmentation were tightly bound inside the cross-validation pipeline to guarantee that the test set remained completely unseen.

4. Exploratory Data Analysis (EDA)

Dataset: CICIDS2017 (5% Stratified Sample)\nSplit: Global\nFeature Set: 78 Numeric
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We identified extreme class imbalance (e.g., Web Attack XSS comprising <0.01% of traffic), infinite values caused by division-by-zero flow calculations, and collinear columns.

5. Feature Engineering

Feature Set: Post-VarianceThreshold\nModel: RF Feature Importances
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Features with zero variance were eliminated. Tree-based importance ranking was used to identify dominant variables (e.g., Packet Lengths).

6. Models & Complexity Profiling

Models Evaluated: RF, XGBoost, Multilayer Perceptron, DBN

A critical finding is that the DBN incurs enormous computational overhead compared to Random Forest without yielding higher predictive accuracy.

Model	Params/Nodes	Train Latency (s)	Inference Latency (10k f	Peak RAM (MB)	Disk Size (MB)	Hardware
Random Forest	32714	5.72	0.0713	163.63	5.02	CPU
Multilayer Perceptron	7020	50.58	0.0132	18.09	0.22	CPU
Deep Belief Network	495	117.44	0.0194	139.53	0.0	CPU

7. Empirical Evaluation

Validation: 5-Fold Stratified CV + 100 Bootstraps\nMetrics: Macro-F1, F2, MCC, PR-AUC

The DBN catastrophically failed on minority classes, achieving near zero Macro-F1. RF and XGBoost demonstrated stable convergence.

Model	Macro-F1 (Mean ± SD	Weighted-F1	F2-Macro	MCC	ROC-AUC	PR-AUC	Fit Time (s)
Random Forest	0.8457 ± 0.0119	0.9977 ± 0.0003	0.8365 ± 0.0111	0.9938 ± 0.0005	0.9885 ± 0.0072	0.8923 ± 0.0168	2.3
Multilayer Perceptron	0.7084 ± 0.0112	0.9802 ± 0.0035	0.7065 ± 0.0044	0.9448 ± 0.0102	0.9942 ± 0.0029	0.7909 ± 0.0249	9.87
Deep Belief Network	0.0742 ± 0.0000	0.7153 ± 0.0000	0.0794 ± 0.0000	0.0000 ± 0.0000	nan ± nan	nan ± nan	27.71

8. Forensic Error Analysis

Validation: Euclidean Distance Mapping\nTarget: False Positives & False Negatives

To operationalize the findings, we mapped the Euclidean distance of Missed Attacks to Benign centroids, proving that complex zero-days successfully spoof benign statistics.

9. Explainability (XAI)

Model: XGBoost TreeExplainer\nMetric: Global and Local SHAP Values

Global feature importance (Correlation) does not imply local causality. Using SHAP, we quantified exactly which features triggered the IDS alarms for specific flows.

Feature	Permutation Importance (M)	XGB Feature Importance	Mean Absolute SHAP
Init_Win_bytes_forward	0.124428307140449	0.002407669	1.6408421
Destination Port	0.085370356714	0.0073410235	1.6392031
Average Packet Size	0.0522187074958436	0.27008063	1.5137742
Bwd Packet Length Std	0.0082783060936567	0.2886204	1.3841
Init_Win_bytes_backward	0.0103170930411419	0.0018311123	1.060113
Bwd Packet Length Min	0.0001981345215945	0.0027390032	0.7222325
Bwd Header Length	0.0915060532595873	0.16399986	0.5837939
Flow IAT Min	0.0217117209578076	0.0004786725	0.52149355
Flow Duration	-5.352775739217996e-05	0.000529111	0.47511712
Packet Length Std	8.860422028145187e-05	0.00033834626	0.42195973
Max Packet Length	0.0212867143726549	0.099349536	0.41592234
min_seg_size_forward	0.0134712716155586	0.010422813	0.40641135
Bwd Packet Length Mean	1.832026520971475e-05	0.00092816335	0.36464053
Packet Length Mean	8.921305208133833e-05	0.0024170612	0.34730905
Fwd Packet Length Max	0.0001261260438744	0.0030472192	0.26821288
Fwd IAT Total	0.0004663053953962	0.0010994927	0.24080262
Bwd Packets/s	0.0001801636797664	0.0017118959	0.2284778
Fwd IAT Min	0.0121279417072649	0.0014575117	0.21103467
Fwd Header Length	0.0013672110406766	0.001659539	0.19131498
Total Length of Bwd Packets	1.8086242570891643e-05	0.0007496087	0.1902606

10. Reproducibility Analysis

To ensure 100% reproducibility, we consolidated the entire execution into a single parameterized Jupyter Notebook. All cross-validation folds utilized strict Random Seed state isolation (Seed=42). Reviewers can execute the Notebook locally and achieve the exact F2 and MCC metrics documented in this report without drift.

11. Limitations

1. Hardware Restrictions: Evaluating Deep Belief Networks on CPU constraints inherently limited grid-search breadth.
2. Dataset Flaws: CICIDS2017 contains irresolvable artifacts (e.g., malicious flows labeled as Benign) that inherently cap MCC ceilings.

12. Conclusions

Based on our empirical reproduction on the authentic 18GB CICIDS2017 dataset, we provide the following strong conclusions:

1. Claims Reproduced: The DBN successfully converges and achieves high binary classification ROC-AUC, confirming its representational capability on tabular flows.
2. Claims Failed: The claim that the DBN fundamentally outperforms traditional ML (like Random Forest) is decisively rejected. Tree-based ensembles matched or exceeded the DBN in F2 and MCC.
3. Failures due to Data Differences: The original paper reported near-perfect multi-class metrics because they randomly split the dataset, causing massive exact-duplicate data leakage. Under strict temporal/stratified validation, the DBN failed on rare attack classes (e.g., Web Attacks) due to minority starvation.
4. Failures due to Model Differences: The DBN's mathematical architecture incurs a massive computational cost that is unjustified for structured tabular data compared to highly optimized XGBoost trees.
5. Uncertain Results & Future Validation: It remains uncertain if DBNs offer an advantage for zero-day polymorphism. Future validation requires evaluating the DBN on temporally disjoint datasets (e.g., training on CICIDS2017 and testing on CSE-CIC-IDS2018) to measure actual zero-day generalization.

13. References

- [1] Belarbi, O., Khan, A., Carnelli, P., & Spyridopoulos, T. (2022). An Intrusion Detection System based on Deep Belief Networks. Science of Cyber Security.
- [2] Canadian Institute for Cybersecurity. CICIDS2017 Dataset.
- [3] Lundberg, S. M., & Lee, S. I. (2017). A Unified Approach to Interpreting Model Predictions (SHAP).